

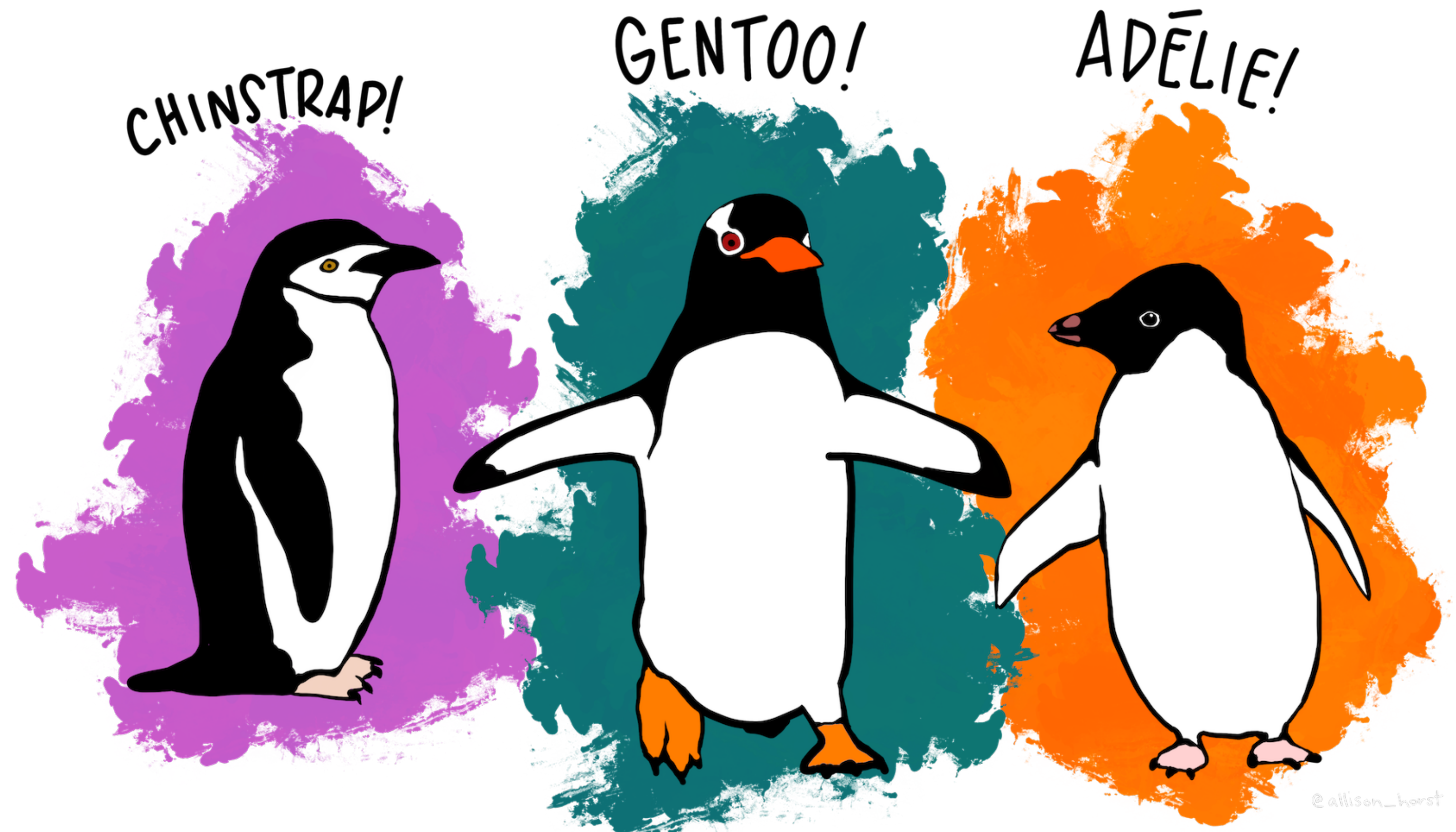
Binary Linear Classifier

DATA 4620: Foundations and Applications of Deep Learning

Instructor: Jonathan Ventura

Palmer Penguins dataset

- Collected by Dr. Kristen Gorman, Palmer Station, Antarctica

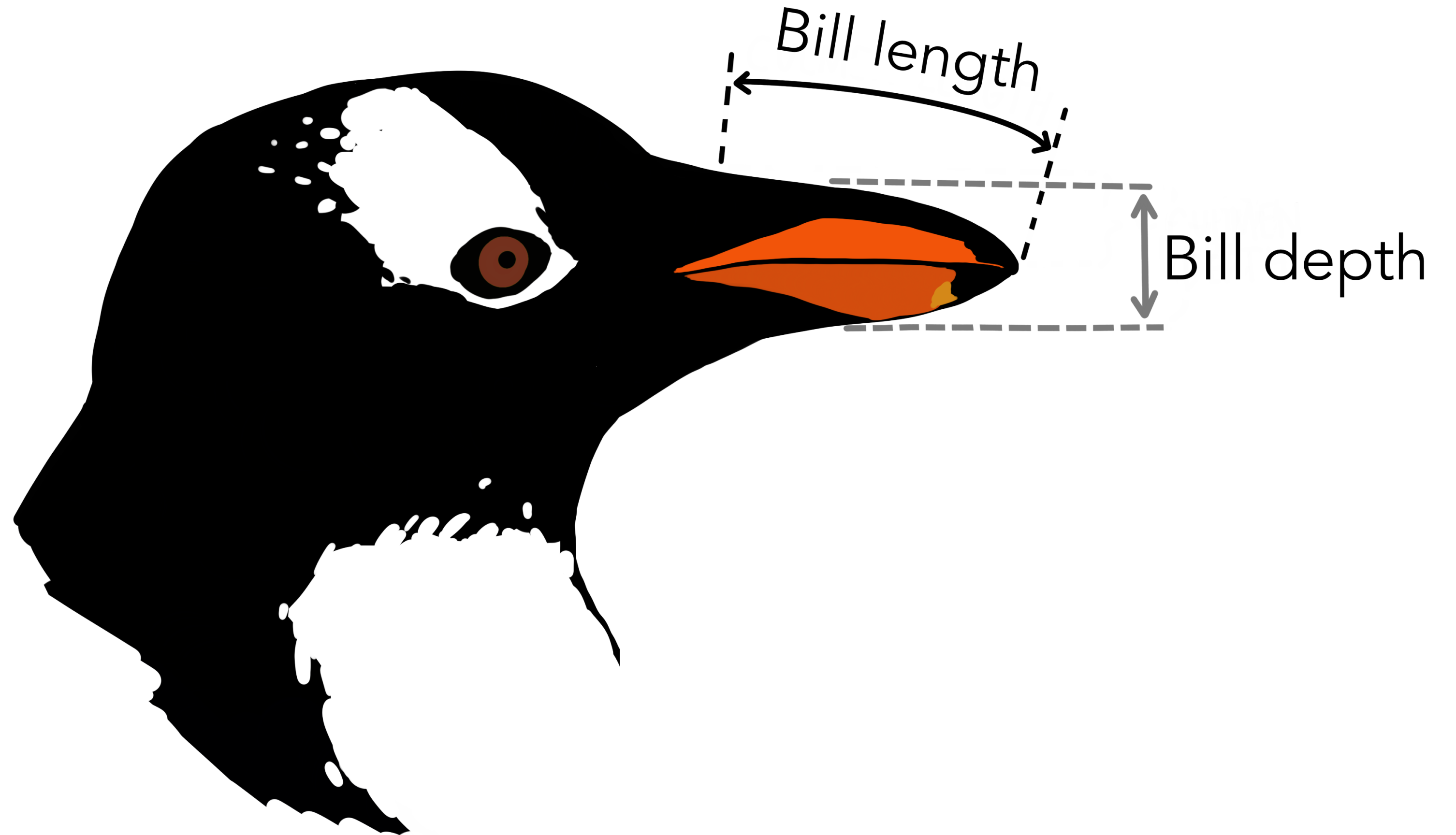


<https://github.com/allisonhorst/palmerpenguins>

<https://www.kaggle.com/parulpandey/palmer-archipelago-antarctica-penguin-data>

Palmer Penguins Dataset

- Contains many measurements of penguins including species, flipper and bill length



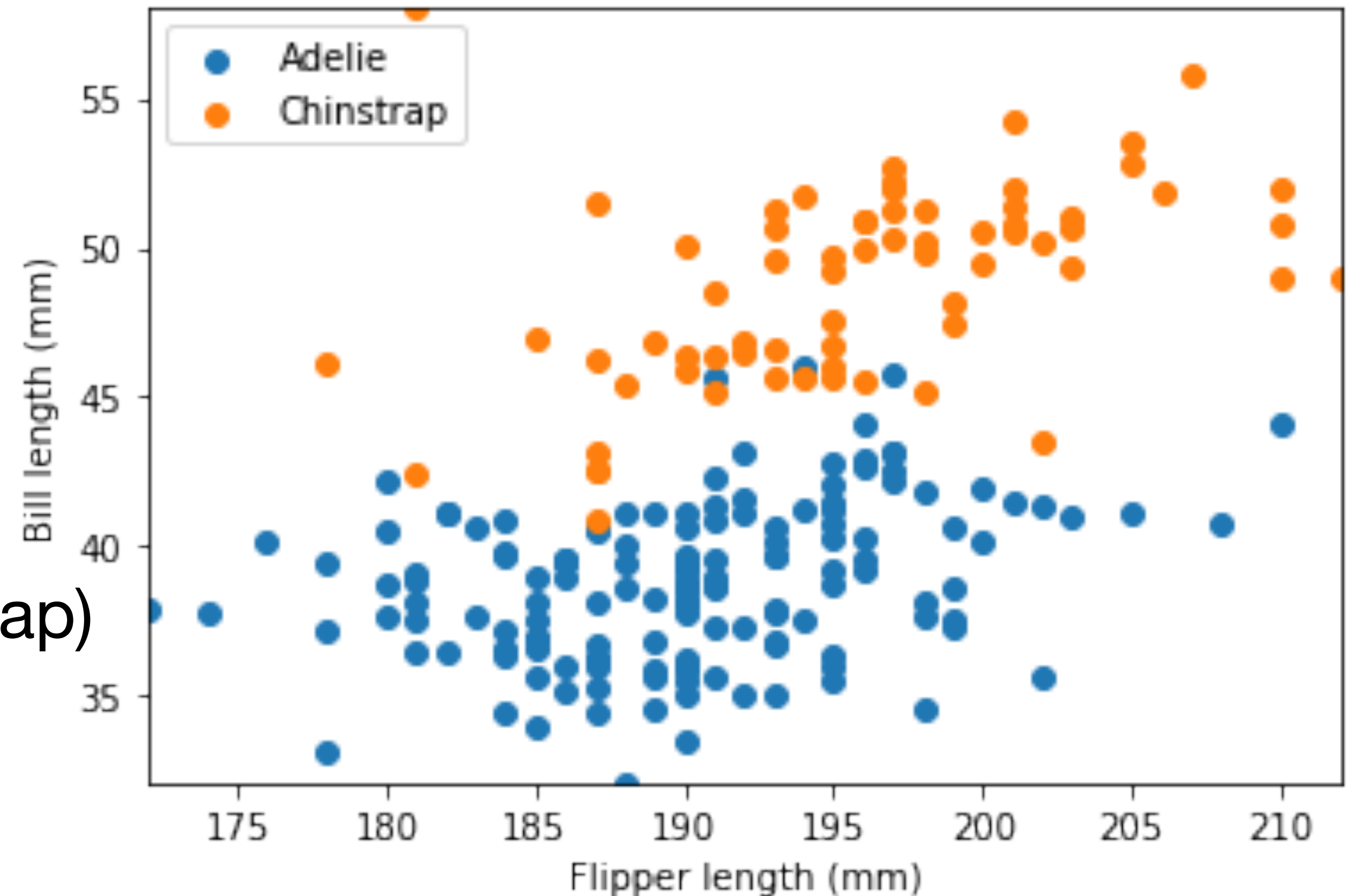
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Palmer Penguins dataset

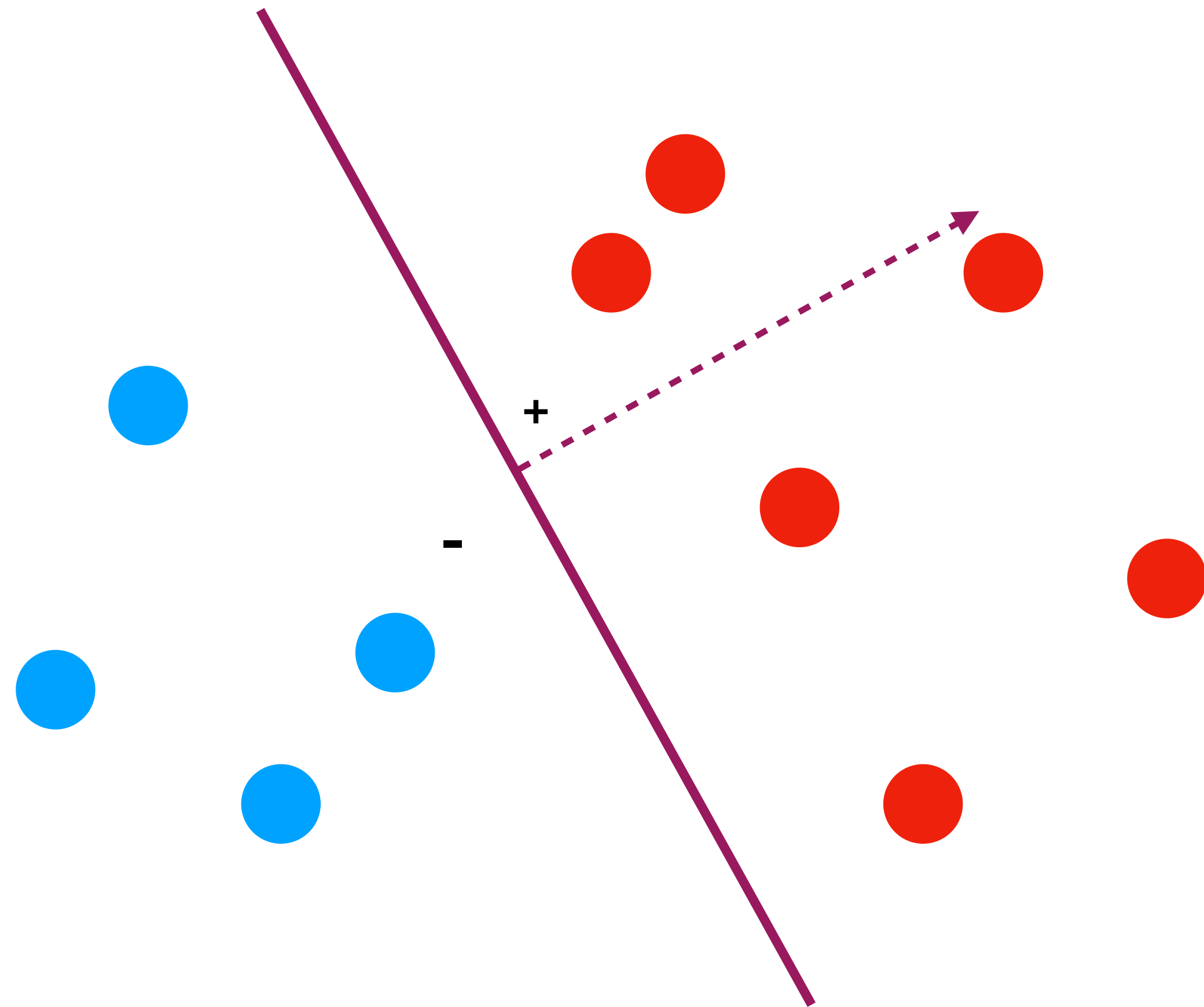
- Goal: classify penguins based on their flipper and bill length

- Features: flipper length (x-axis)
bill length (y-axis)
- Label: species (Adelie or Chinstrap)



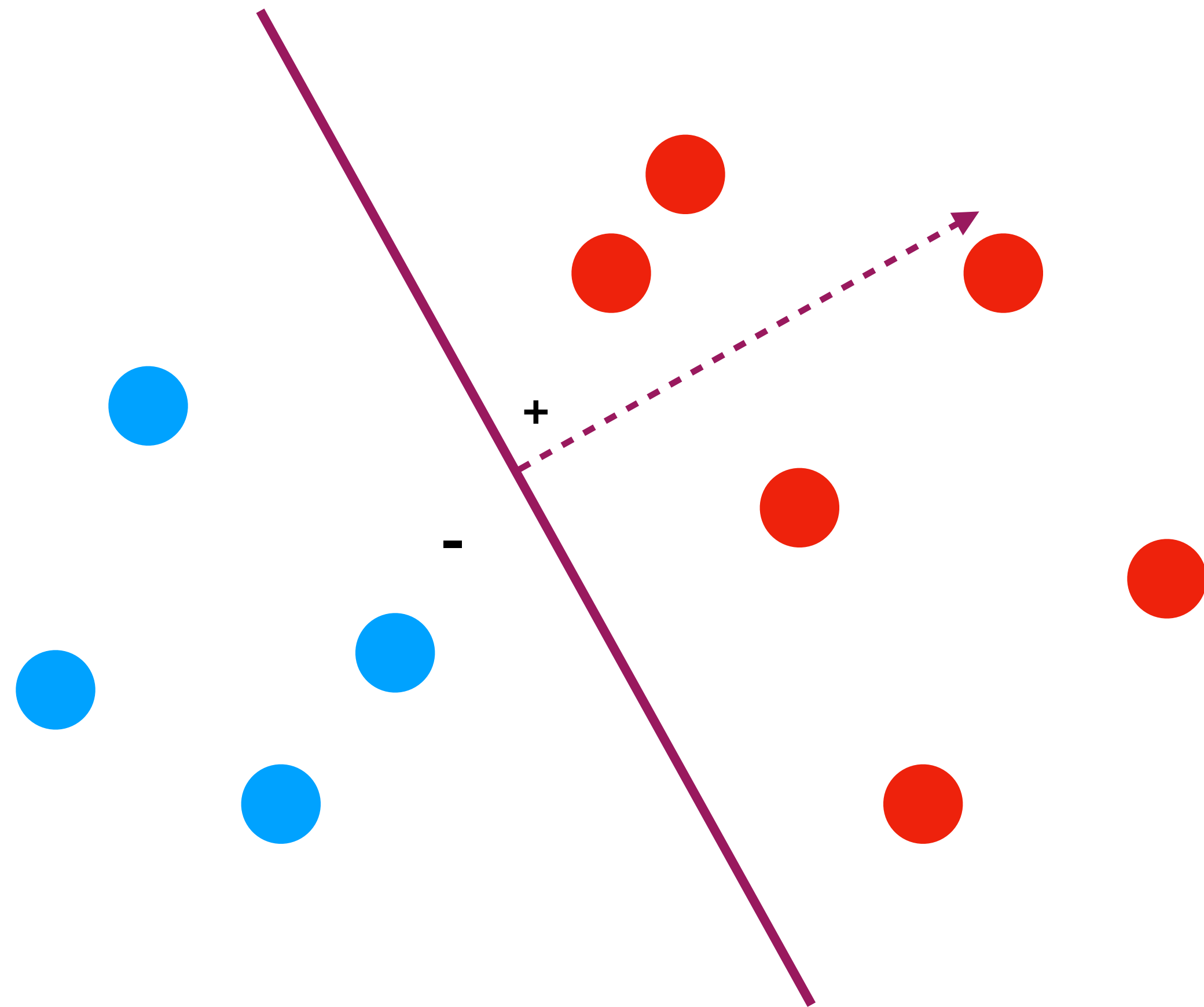
Binary linear classifier

- A binary linear classifier divides space into a “positive” class and “negative” class



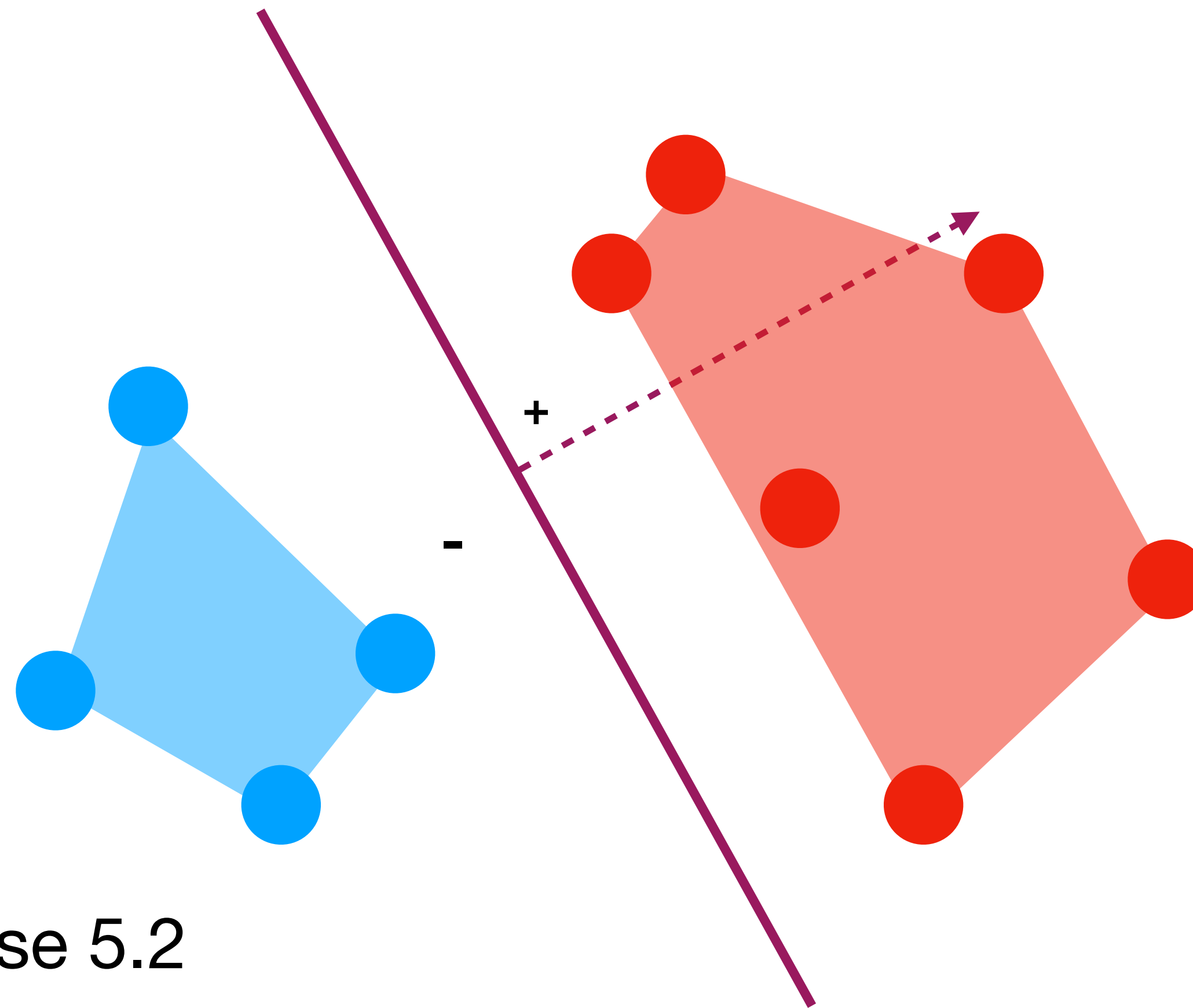
Binary linear classifier

- When is data linearly separable? When is it not?



Binary linear classifier

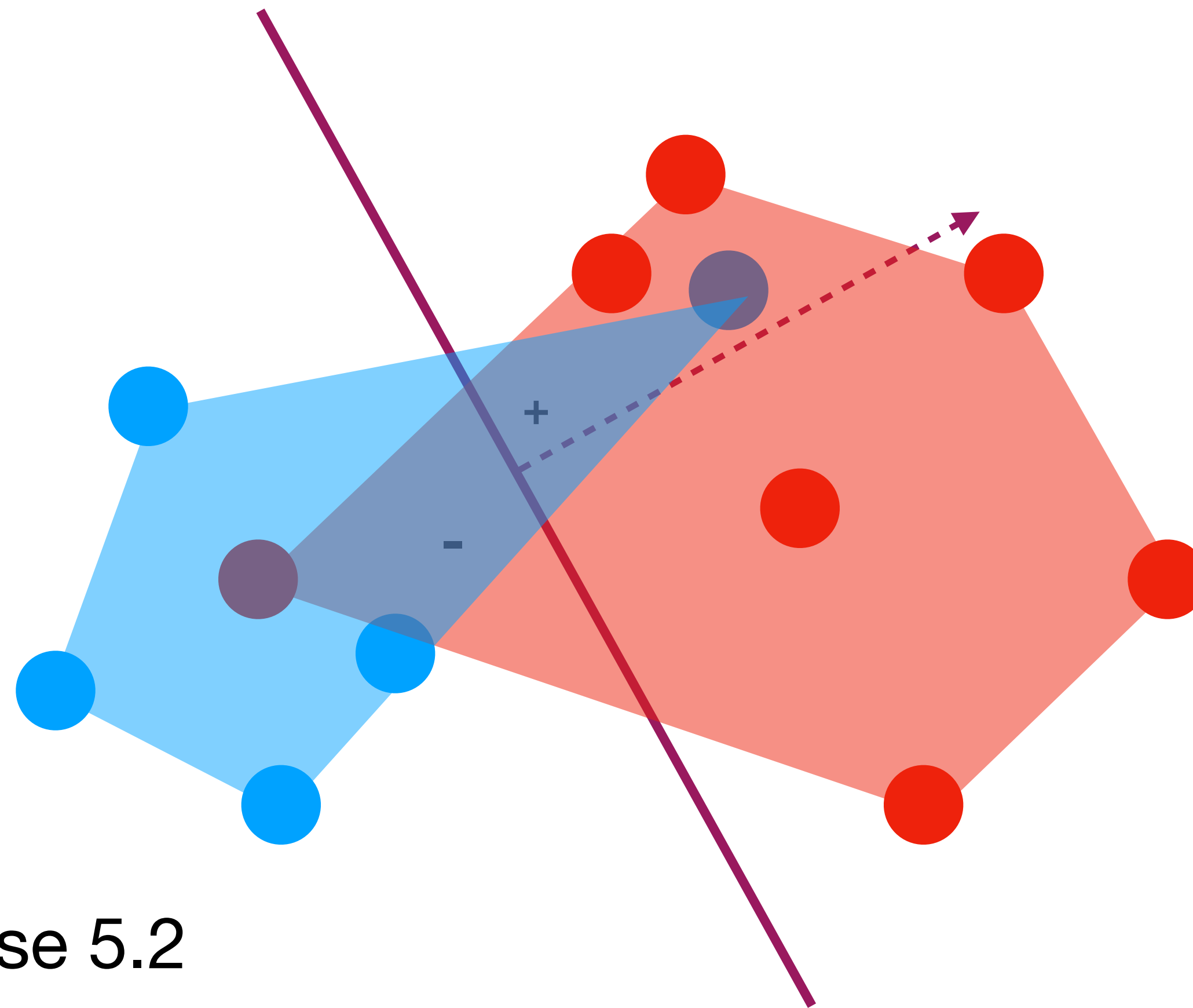
- Points are linearly separable when their *convex hulls* do not intersect



- See Bishop Exercise 5.2

Binary linear classifier

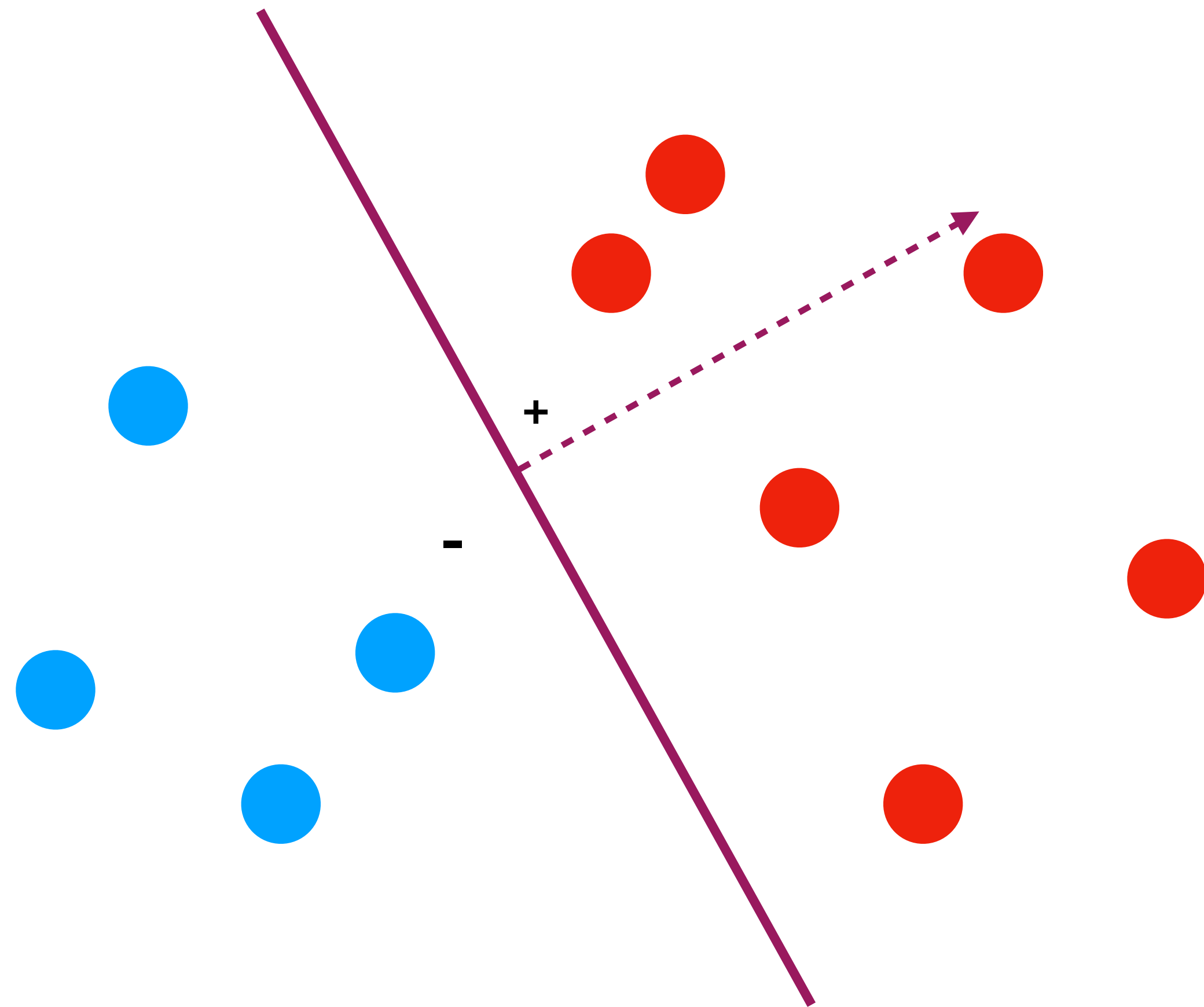
- Points are linearly separable when their *convex hulls* do not intersect



- See Bishop Exercise 5.2

Binary linear classifier

- Is there a unique solution to binary linear classification?



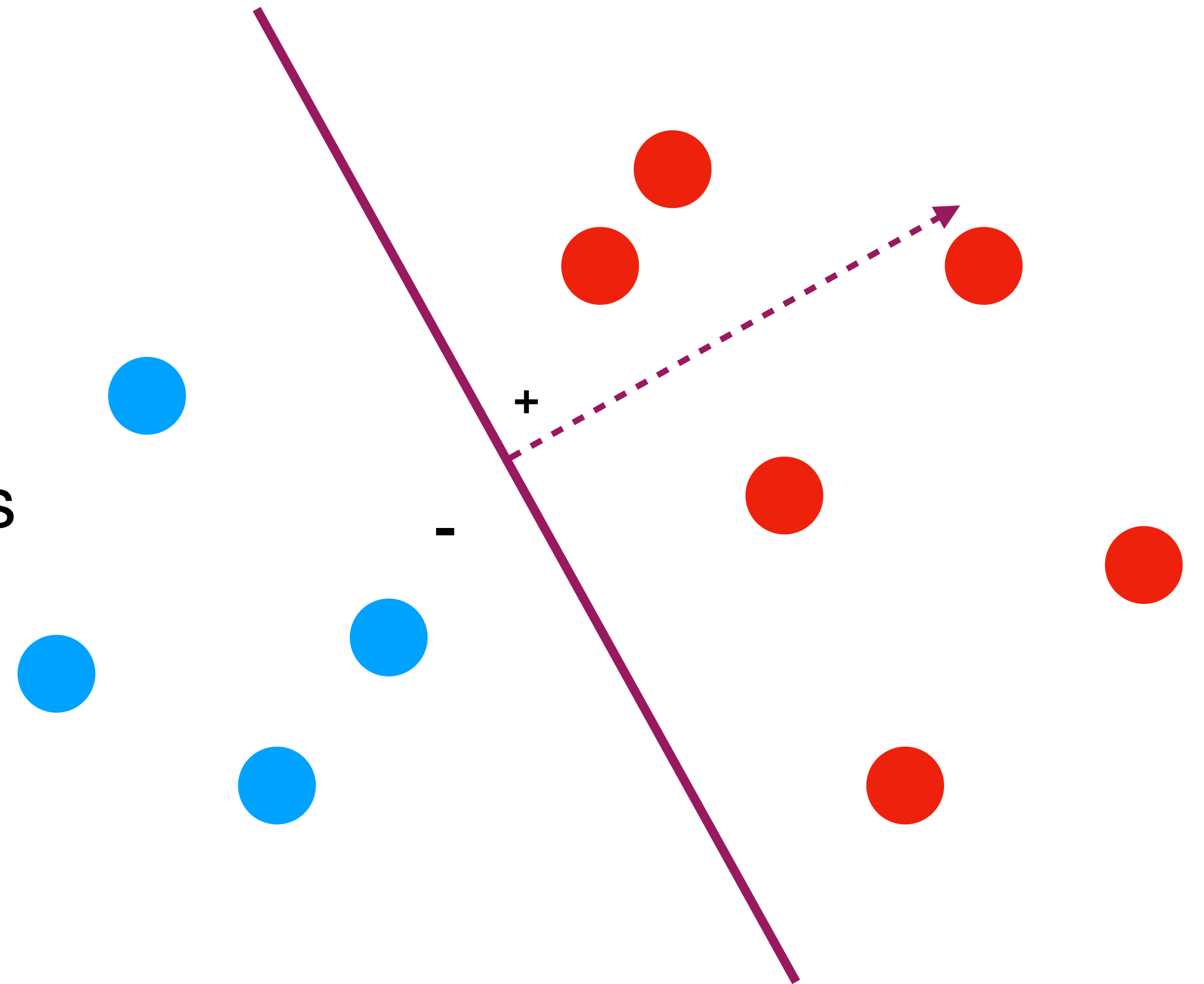
Linear classifier divides space with a hyperplane

- A linear classifier divides space into a “positive” class and “negative” class

$$z = w_1x_1 + w_2x_2 + b$$

$$z = \mathbf{w}^T \mathbf{x} + b$$

- $\mathbf{w}$ is a vector of “weights” that determines the line’s orientation
- b is a “bias” value that translates the line along the normal direction

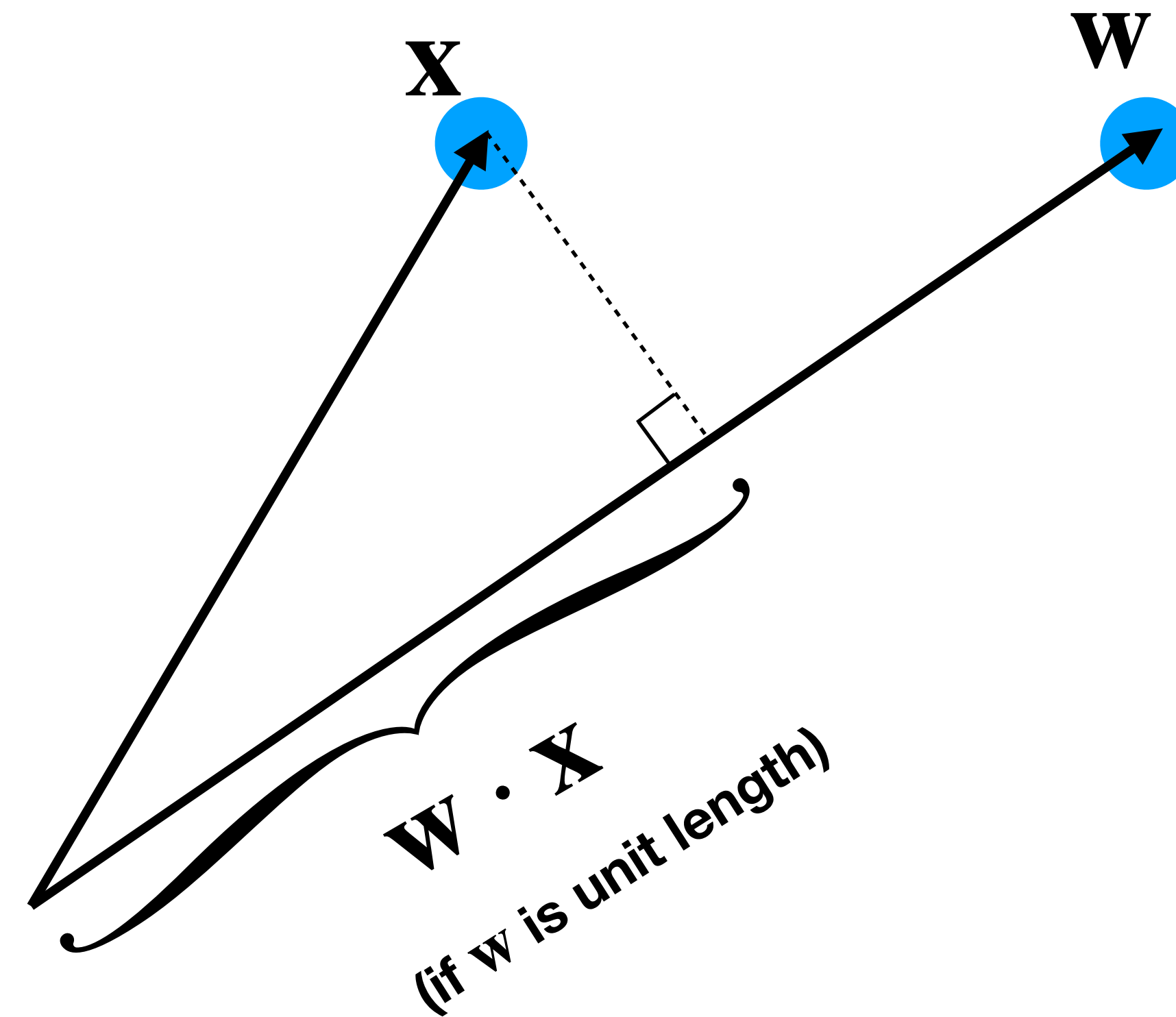


Dot product

- ...or “inner product”

$$\mathbf{w}^T \mathbf{x} \equiv \mathbf{w} \cdot \mathbf{x}$$

- What does it do?



- What does the bias do?

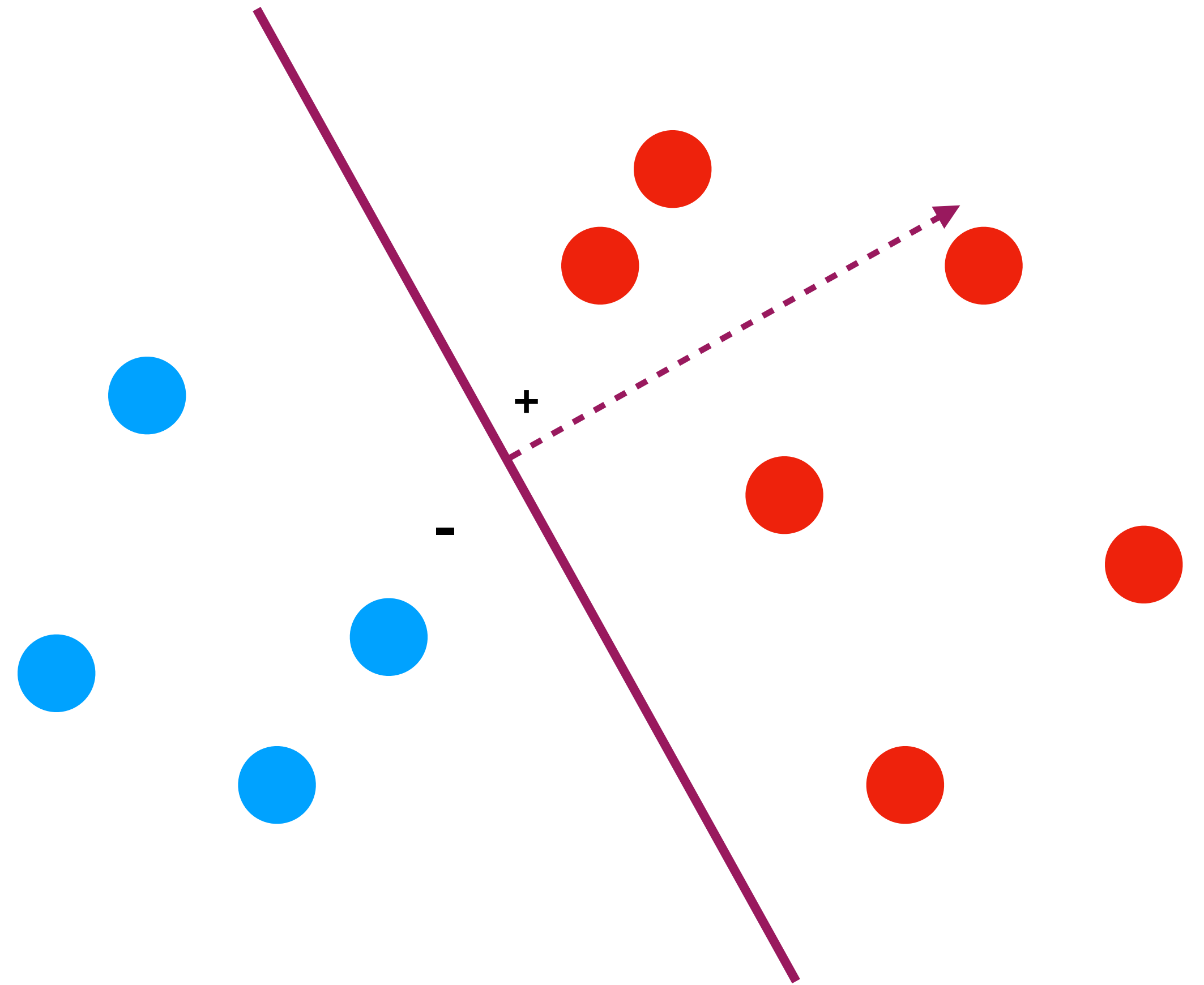
$$\mathbf{w}^T \mathbf{x} + b$$

Interpretation of weights

- How to interpret the weights of the classifier?

$$z = w_1x_1 + w_2x_2 + b$$

- Sign of weight: direction of relationship
- Magnitude of weight: strength of relationship



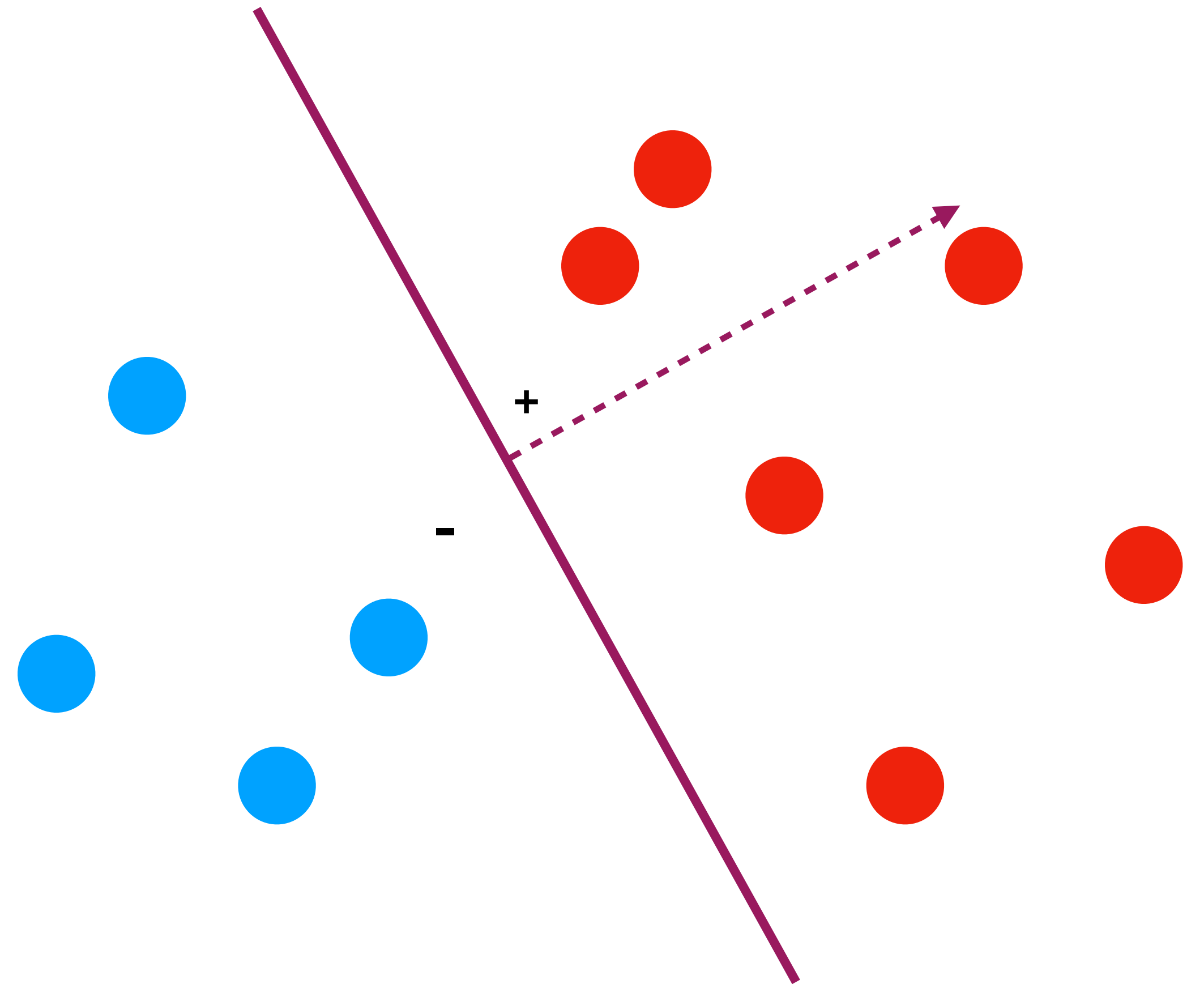
“Hiding” the bias term

- We can “hide” the bias term by adding a 1 onto the end of the data point:

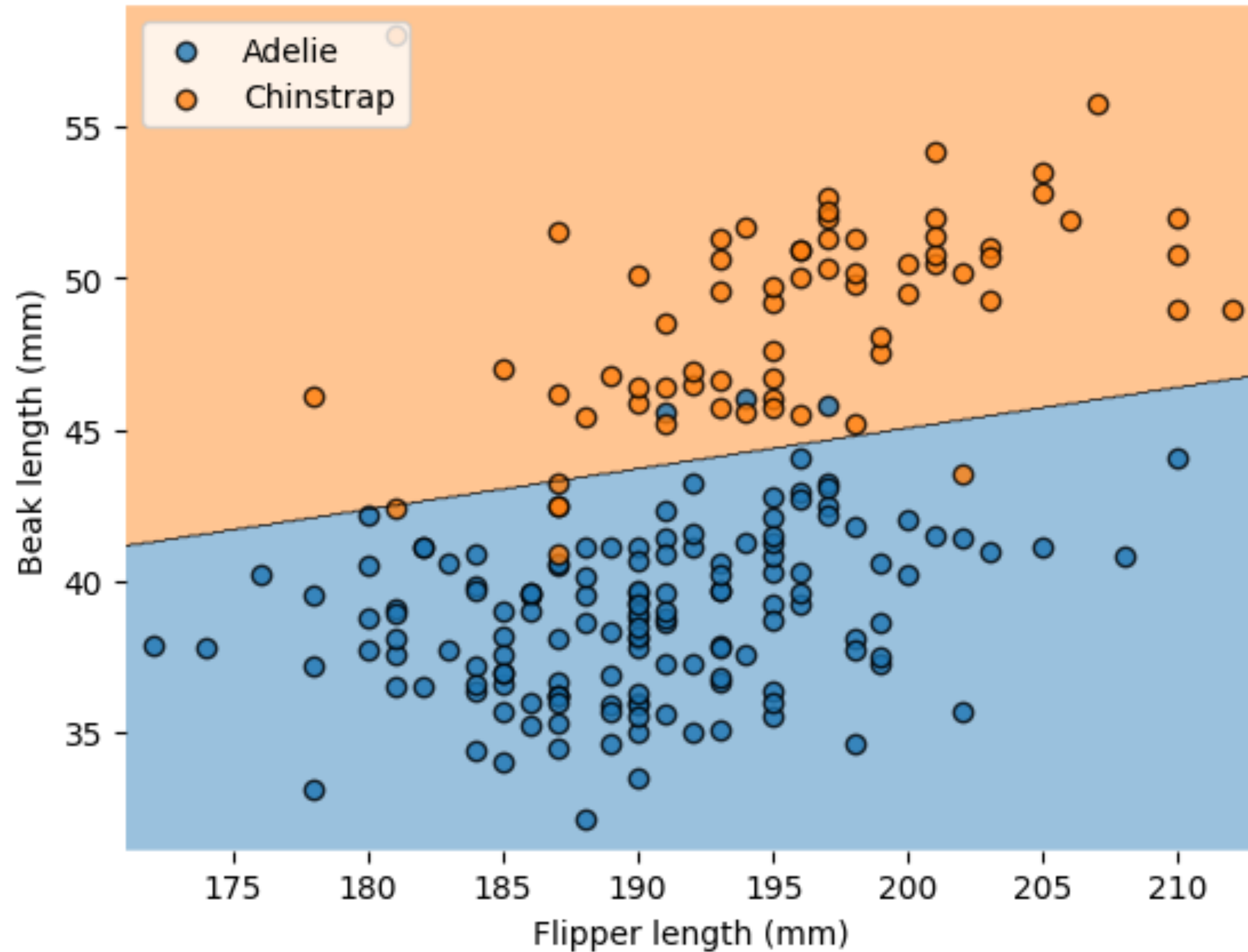
$$z = w_1x_1 + w_2x_2 + b1$$

$$z = \begin{bmatrix} \mathbf{w}^T & b \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}$$

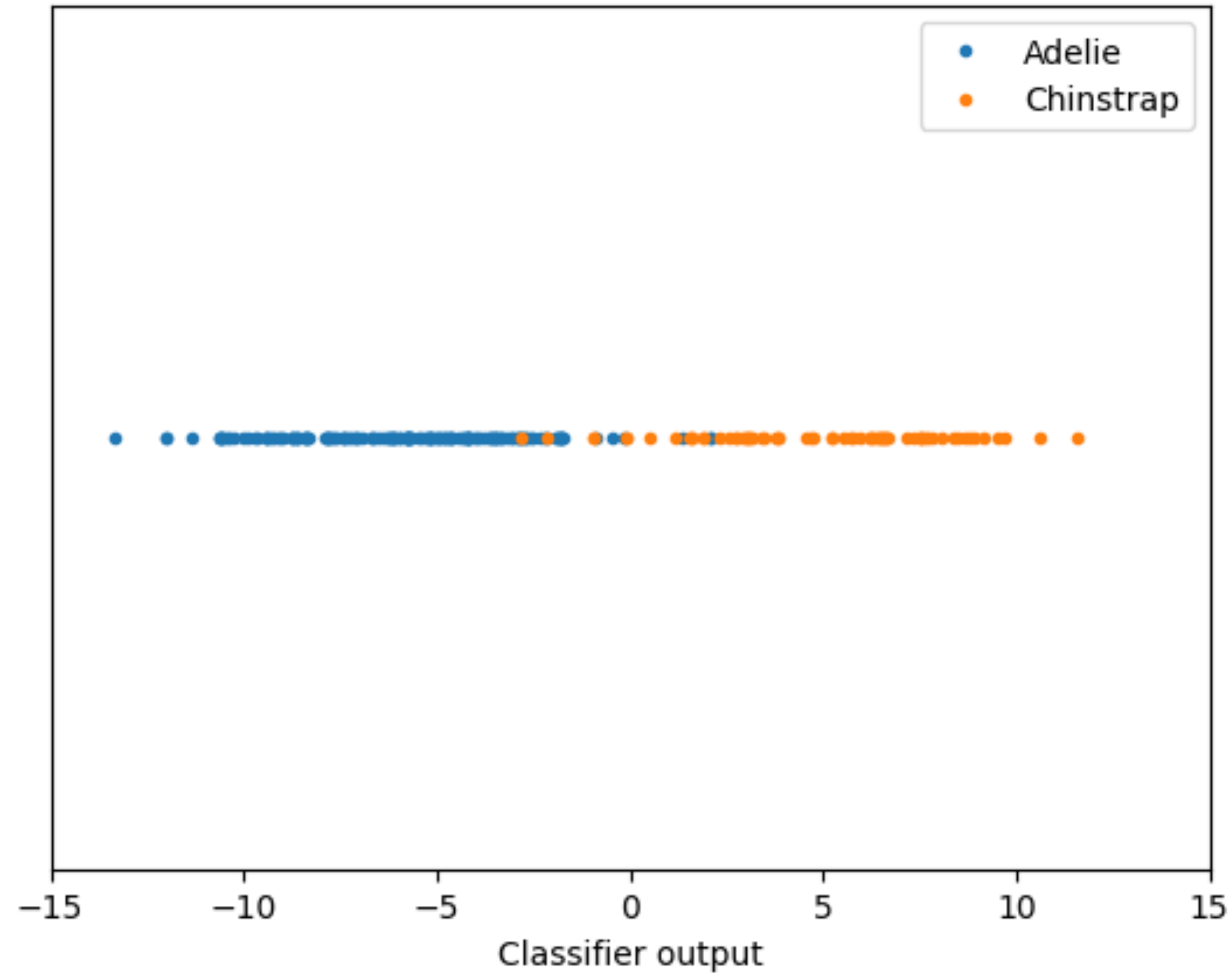
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Binary classifier result



Binary classifier result



Linear classifier by hand

- Problem 1:
 - Find a solution for the linear classifier for the following points:
 - Positive points: $(1,1)$
 - Negative points: $(-1,-1)$

Linear classifier by hand

- Problem 2:
 - Find a solution for the linear classifier for the following points:
 - Positive points: $(1,1)$ and $(-1,1)$
 - Negative points: $(-1,-1)$ and $(1,-1)$

Linear classifier by hand

- Problem 3:
 - Find a solution for the linear classifier for the following points:
 - Positive points: $(1,1)$ and $(-1,-1)$
 - Negative points: $(-1,1)$ and $(1,-1)$